

TASK 2: EXPLORATORY DATA ANALYSIS (EDA)

Wine Chemistry Dataset — Structured Analysis & Insights

Project	Exploratory Data Analysis
Dataset	Wine recognition dataset (scikit-learn / UCI)
Records	178 samples
Features	13 numeric features + target
Classes	3 wine classes
Analysis tools	Python • Pandas • Matplotlib • SciPy

Executive summary: This report follows a complete EDA workflow: defining questions, checking structure and quality, studying distributions, finding relationships and anomalies, testing a hypothesis, and converting the results into actionable insights.

Prepared for internship Task 2 submission

1. Objectives & Dataset Understanding

EDA questions

1) How is the dataset structured? 2) Are there missing values or duplicate records? 3) Which features vary most? 4) Which variables are strongly related? 5) Do wine classes show statistically meaningful differences in chemical measurements? 6) Which observations may require attention as potential outliers?

Dataset overview

The Wine dataset contains chemical measurements for wines belonging to three classes. The analysis uses the 13 continuous chemical features and the class label. No target leakage is introduced into the descriptive statistics of the features.

Metric	Result
Rows	178
Numeric feature columns	13
Target classes	3
Missing values	0
Duplicate rows	0
Class 0 / 1 / 2	59 / 71 / 48

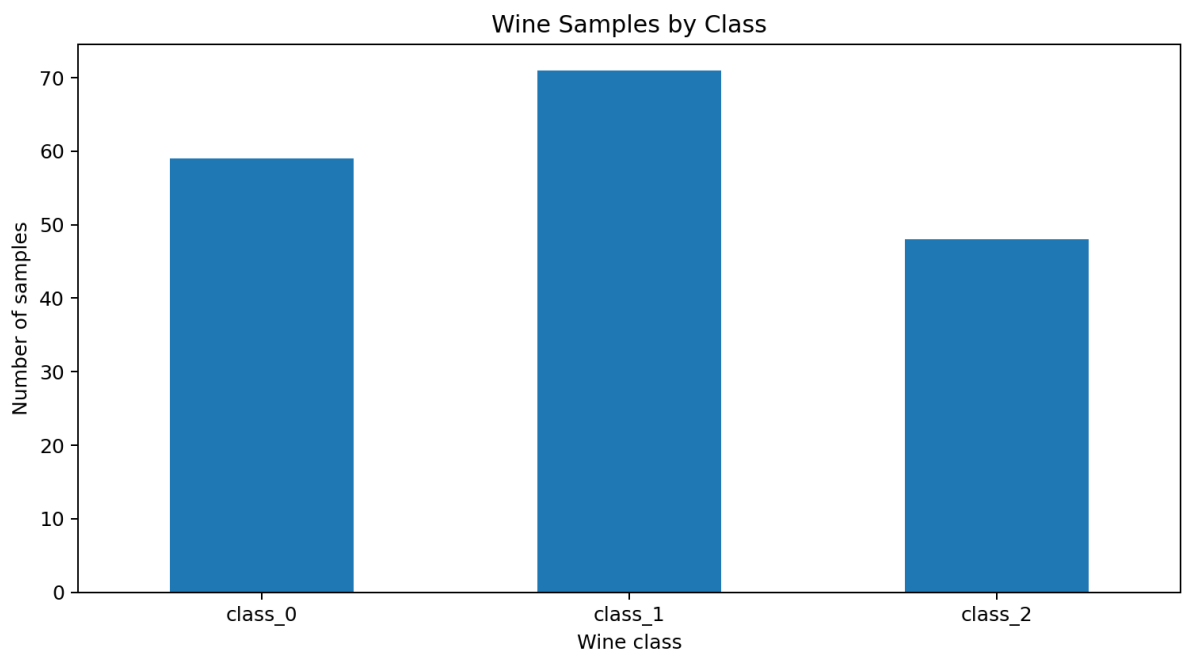


Figure 1. The three classes are not perfectly balanced; class 1 is the largest group.

2. Data Structure & Descriptive Statistics

All 13 predictors are numeric continuous measurements. The target is categorical with three classes. The table below highlights the central tendency and spread of the main variables.

Feature	Mean	Std. dev.	Min	Median	Max
alcohol	13.00	0.81	11.03	13.05	14.83
malic_acid	2.34	1.12	0.74	1.87	5.80
ash	2.37	0.27	1.36	2.36	3.23
magnesium	99.74	14.28	70.00	98.00	162.00
total_phenols	2.30	0.63	0.98	2.35	3.88
flavanoids	2.03	1.00	0.34	2.13	5.08
color_intensity	5.06	2.32	1.28	4.69	13.00
hue	0.96	0.23	0.48	0.96	1.71
proline	746.89	314.91	278.00	673.50	1680.00

Quality check: No missing values and no duplicate rows were found. Because the features use different measurement scales, direct comparison of raw magnitudes should be avoided when modelling; standardization would be appropriate for scale-sensitive algorithms.

3. Distributions, Trends & Patterns

Distribution analysis helps reveal class separation, skewness, concentration and unusual observations. Alcohol is visibly distributed differently across the three wine classes.

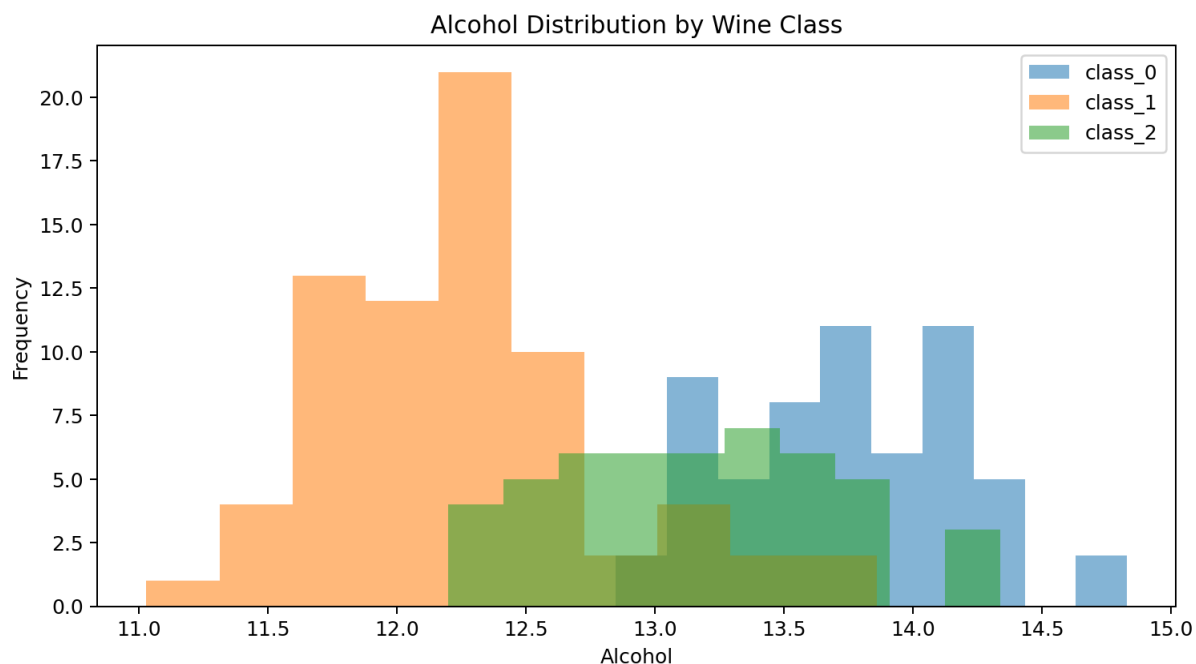


Figure 2. Alcohol values show substantial differences between the wine classes.

Pattern: Class 1 tends to have lower alcohol values than Class 0, while Class 2 lies between them. This suggests alcohol may carry useful information for distinguishing classes, although it should not be used alone.

4. Relationships Between Variables

Bivariate analysis identifies features that move together and may contain overlapping information. The strongest positive correlation is between total phenols and flavanoids.

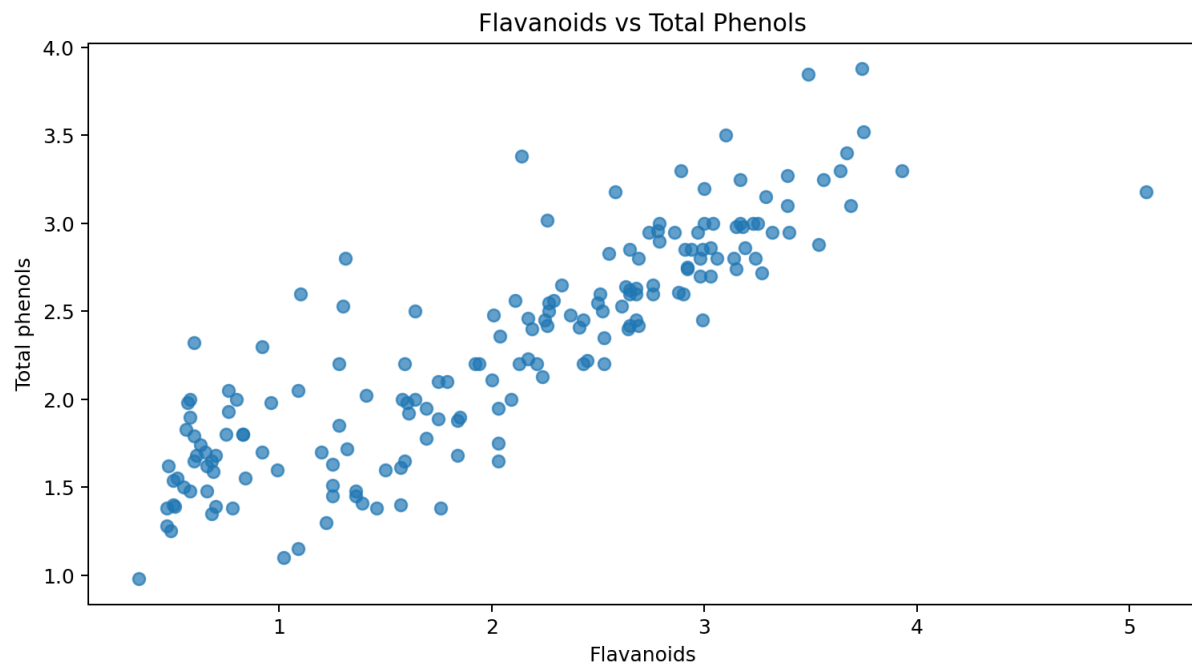


Figure 3. Total phenols and flavanoids show a strong positive relationship.

Rank	Feature pair	Pearson r
1	total_phenols ↔ flavanoids	0.865
2	flavanoids ↔ od280/od315_of_diluted_wines	0.787
3	total_phenols ↔ od280/od315_of_diluted_wines	0.700
4	flavanoids ↔ proanthocyanins	0.653
5	alcohol ↔ proline	0.644
6	total_phenols ↔ proanthocyanins	0.612

5. Correlation Analysis

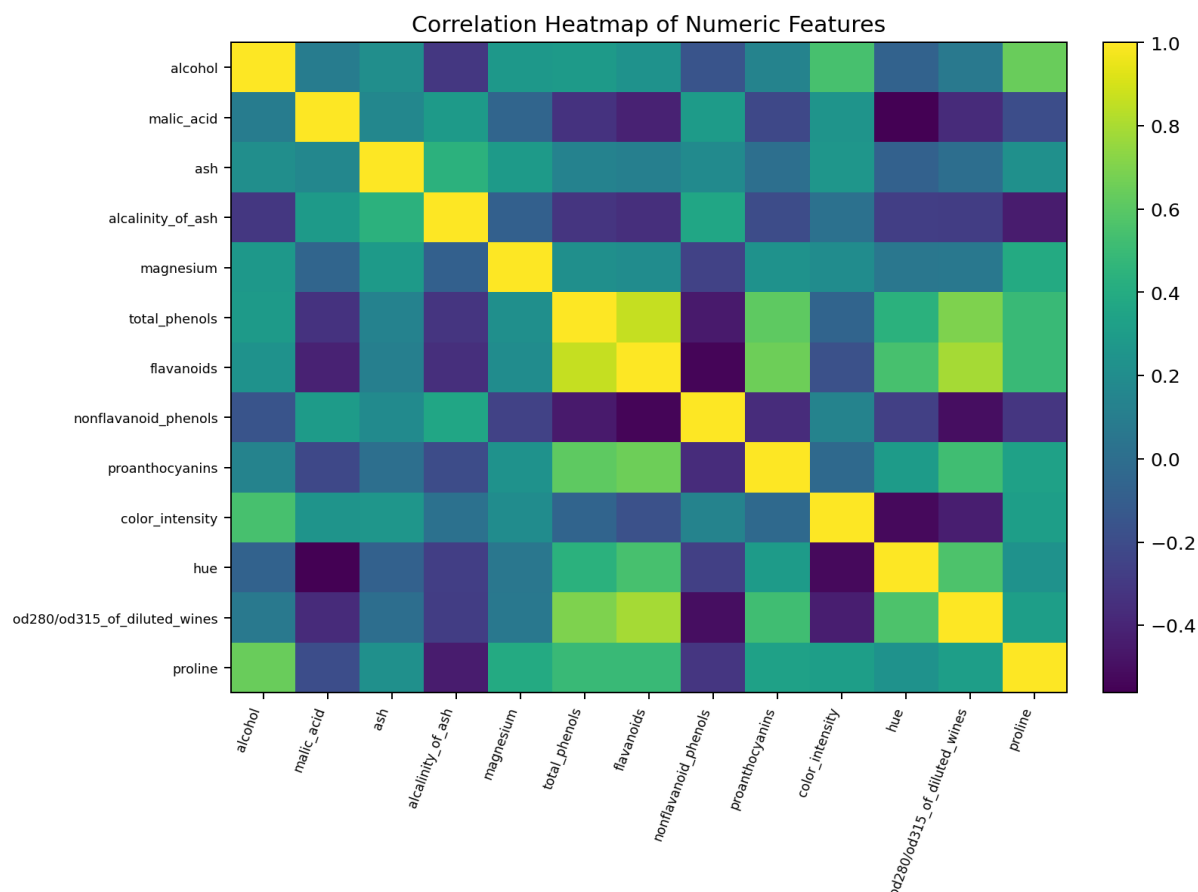


Figure 4. The heatmap summarizes linear relationships among all 13 numeric features. The strongest relationship is total phenols vs flavanoids ($r = 0.865$).

Interpretation: Strongly correlated variables may provide overlapping information. For predictive modelling, this can matter for feature selection and multicollinearity-sensitive methods. Correlation, however, does not establish causation.

6. Anomaly & Outlier Detection

Potential outliers were screened using the $1.5 \times \text{IQR}$ rule. This is a screening technique, not proof that a record is erroneous; extreme chemical measurements may be legitimate observations.

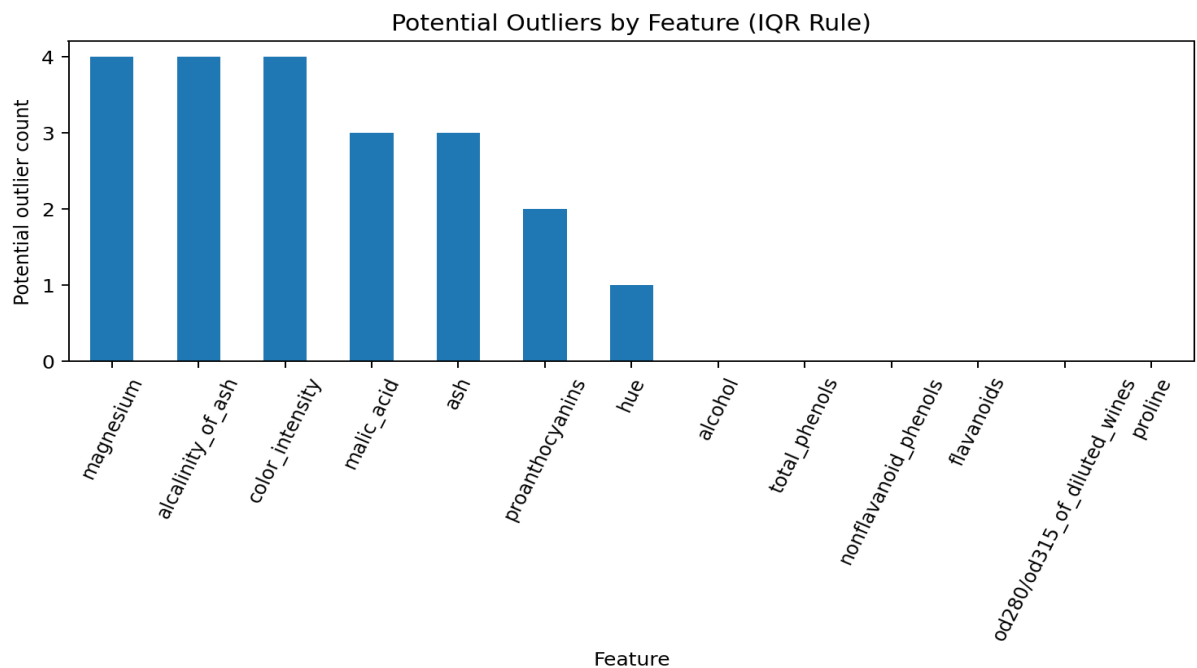


Figure 5. Potential outlier counts identified by the IQR rule.

The largest counts occur in alkalinity_of_ash, magnesium and color_intensity (4 potential outliers each). Before deleting any such records, their source and domain meaning should be checked. For this report, no observations were removed.

7. Hypothesis Testing & Validation

Question: Is the mean alcohol content the same across all three wine classes?

H₀: The three class means are equal.

H_a: At least one class mean differs.

Test	Statistic	p-value	Decision ($\alpha = 0.05$)
One-way ANOVA	F = 135.08	3.32e-36	Reject H ₀

The ANOVA result is highly significant (F = 135.08, p = 3.32e-36), so the null hypothesis is rejected. This provides statistical evidence that alcohol levels are not equal across all three classes. A post-hoc test would be the next step to identify exactly which class pairs differ.

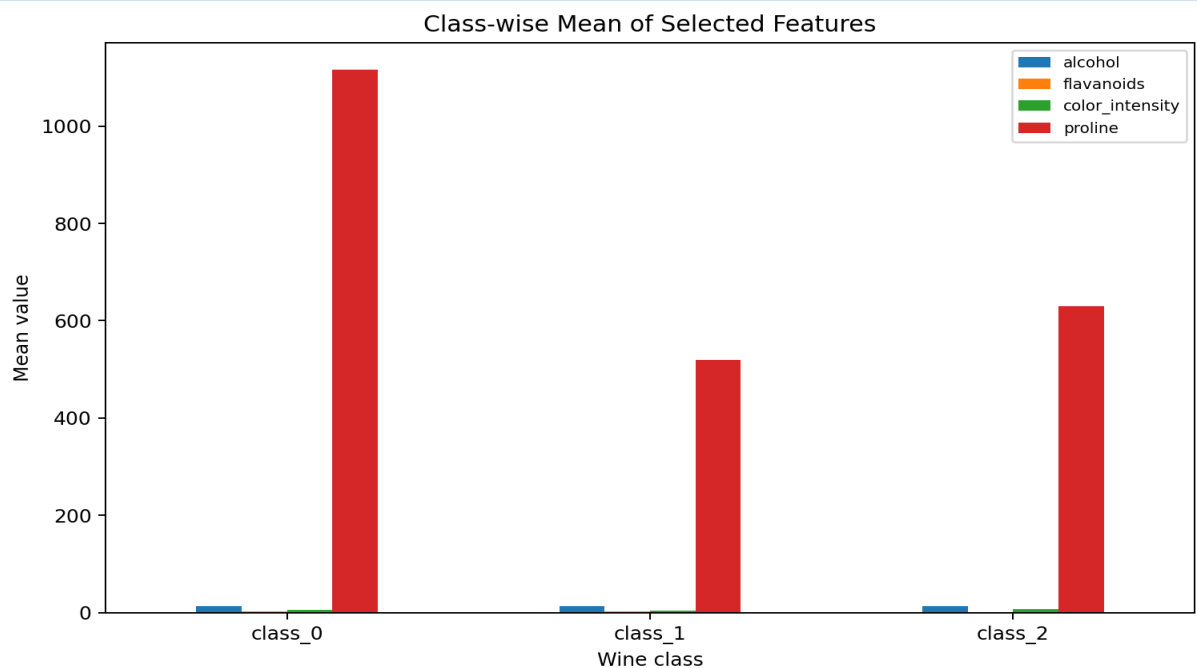


Figure 6. Class-wise means for selected chemical features.

8. Key Findings & Recommendations

Key findings

- The dataset is compact and analysis-ready: 178 records, 13 numeric predictors and 3 classes.
- No missing values or duplicate records were detected.
- Alcohol shows clear class-wise differences; ANOVA strongly rejects equal means.
- Total phenols and flavanoids have a strong positive correlation ($r \approx 0.865$).
- Several variables contain IQR-flagged observations, but none were automatically deleted.
- Different feature scales suggest standardization before scale-sensitive machine-learning models.

Recommendations for further analysis

1. Perform post-hoc pairwise tests after ANOVA.
2. Standardize features and compare classification models such as Logistic Regression, KNN and SVM.
3. Use PCA to study dimensionality and visualize class separation.
4. Investigate outliers against domain/source information before treatment.
5. Evaluate model performance with stratified cross-validation rather than a single train/test split.

Conclusion: The EDA successfully identifies the dataset structure, quality, distributions, relationships, potential anomalies and a statistically supported class difference. These findings provide a strong foundation for the next stage of predictive modelling.

Dataset reference: scikit-learn's built-in Wine recognition dataset, originally from the UCI Machine Learning Repository.

Appendix: Reproducible Python Workflow

The core analysis can be reproduced with the following Python workflow. The submitted report contains the resulting tables, statistical test and visualizations.

```
# Task 2 - Exploratory Data Analysis
from sklearn.datasets import load_wine
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import f_oneway

wine = load_wine(as_frame=True)
df = wine.frame

# Structure and quality
print(df.shape)
print(df.info())
print(df.isna().sum())
print(df.duplicated().sum())

# Descriptive statistics
print(df.describe())

# Class distribution
print(df["target"].value_counts())

# Correlation
corr = df.drop(columns="target").corr()
print(corr)

# Example hypothesis test: alcohol across classes
groups = [df.loc[df.target == c, "alcohol"]
          for c in sorted(df.target.unique())]
F, p = f_oneway(*groups)
print("F =", F, "p =", p)

# Outlier screening using IQR
for col in wine.feature_names:
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1
    mask = (df[col] < q1 - 1.5*iqr) | (df[col] > q3 + 1.5*iqr)
    print(col, "potential outliers:", mask.sum())
```